



ARTICLE



<https://doi.org/10.1057/s41599-026-06989-4>

OPEN

Algorithmic human resource management as a mode of algorithmic governance: transparency, fairness, and human agency in the digital workplace

Zhisheng Chen¹✉

The rapid digitalization of the workplace is reconfiguring the foundations of human resource management (HRM). This paper conceptualizes algorithmic HRM as a business process technology that transforms traditional people management practices through automation, predictive analytics, and workflow redesign. By comparing algorithmic and conventional HRM processes across six core domains—workforce planning, recruitment, training, performance management, compensation, and employee relations—the study demonstrates how algorithms shift HRM from intuition-driven decisions to data-based and adaptive systems. While these transformations offer efficiency gains, personalization, and new avenues for strategic value creation, they also raise critical concerns about transparency, fairness, privacy, and employee autonomy. The paper argues that algorithmic HRM represents both a technological and organizational frontier in the future of work, demanding hybrid governance approaches, ethical safeguards, and the cultivation of human capabilities that algorithms cannot replace. In doing so, it highlights implications for managers, workers, and policymakers navigating the challenges of an algorithmically mediated employment landscape.

¹Wuxi Taihu University, Wuxi, China. ✉email: njuczs@nuaa.edu.cn

Introduction

Technology advancements, including artificial intelligence, virtual reality, and voice recognition, are becoming more prevalent during the fourth wave of industrialization (Shank et al. 2019). These cutting-edge technologies have strongly affected work practices, worker engagement, and workflows locally or worldwide (Abraham et al. 2019). While analyzing four decades of technical progress in the digital realm, Bondarouk et al. (2017) pointed out that quick technology advancements have given rise to a fresh, perceptive digital environment.

However, some businesses primarily gather personnel information to describe employee retention, compensation expenses, or employee makeup and rarely use predictive and prescriptive HRM algorithms (Marler and Boudreau 2017; Meijerink and Bondarouk 2023). These styles of algorithmic HRM contribute to a large extent to the achievement of strategic organizational goals. People foresee algorithmic management use as one of the growing trends. Automating HR-related tasks differentiates it from earlier e-HRM programs (Sienkiewicz 2021).

HR manager duties such as employee relations, recruitment, performance management, and payroll can be influenced by or perhaps ruled by computer algorithms. Employee behaviour produces digital traces such as performance logs, workflow metadata, and communication patterns that serve as input for algorithmic HRM systems. These systems analyse such data to inform HR decisions, without necessarily implying autonomous AI-driven operations (Garcia-Arroyo and Osca 2021). HRM issues have been predicted using heuristic-based techniques like neural networks, which operate by identifying non-linear patterns through iterative weight adjustments that approximate cognitive associations (Zhang et al. 2021).

Across various discipline thresholds, research into algorithmic human resource management is gaining traction (Veen et al. 2020). Algorithmic HRM attracts good employees, improves the quality of training, assists in payroll, and conducts practical performance evaluations of workers while cutting back on the time, money, and cognitive biases driven by humans (Pessach et al. 2020; Torres and Mejia 2017). Algorithmic tools are, therefore, essential for improving personnel procedures and operations.

Even though algorithmic HRM has many possible advantages, concerns have also been raised about its negative impacts, including worker exploitation, job intensity, deterioration of perceived justice, and the illusion of objectivity, and ultimately affects the overall business performance and goodwill of the organization (Newman et al. 2020; Vassilopoulou et al. 2022). The intricacy of HR issues, the constraints of limited data sets, responsibility concerns relating to justice, various ethical and legal problems, as well as the possibility of unfavourable employee reactions to management decisions made using data-based algorithms have all been highlighted by researchers as limits that frequently result from the implementation of algorithms in HRM (Tambe et al. 2019).

The application of algorithmic human resource management is an innovative field of research and has not been extensively studied. The extent of use and the effects of a more automated human task management method are poorly understood. While integration around key issues is beginning, there is a tendency to use multiple names to express what appears to be the same subject, leading to a lack of structural clarity that could inhibit academics from building a cohesive knowledge base (Yaniv 2011). In addition to lacking structural coherence, academic research on particular digital HRM activities and the elements that make them up has trailed behind the term's acceptance in the real world. The proliferation of applications in HR practice has dwarfed the growth in academic interest in HR-related algorithms

(Cheng and Hackett 2021). Similarly, (Marler and Boudreau 2017) found that despite more blogs and news coverage, management researchers do not seem to pay much attention to the topic. Theorists and practitioners should take a more active part in algorithmic HRM integration, moving it away from its current goal of improving HRM's efficacy and effectiveness toward focusing on strategically optimal traits and establishing the conditions necessary for these results.

The purpose of this study is to provide a conceptualization of algorithmic HRM by synthesizing and resolving inconsistencies in existing scholarly definitions, and by examining how algorithmic systems reshape key HR functions. Rather than treating algorithmic HRM as a set of isolated tools, this paper conceptualizes it as a mode of algorithmic governance. Accordingly, the study analytically maps the application of algorithmic systems across recruitment, selection, training, performance management, and employee relations, identifying both the advantages such as efficiency, consistency, and scalability and the risks, including bias, opacity, and reduced autonomy.

This paper makes three contributions. First, it clarifies the core construct of algorithmic HRM by integrating insights from information systems, HRM, and organizational governance. Second, it provides a structured analytical framework that explains how algorithms operate across HR activities and how their dual effects of beneficial and detrimental emerge in practice. Third, it positions algorithmic HRM within the debate on algorithmic governance, illustrating how digital decision systems reshape managerial rationality and the dynamics of power, fairness, and human agency in the workplace.

The remaining chapters are distributed as follows. The study begins with a discussion of the background of algorithmic HRM. The second part analyses the opportunities for algorithmic HRM practice. Further, the study discusses algorithmic HRM strengths, weaknesses, and measures. Finally, future research directions are suggested.

Background

Algorithmic Management. An algorithm is a computational formula that uses statistical models or predefined decision rules to perform tasks autonomously, often without direct human input (Duggan et al. 2020). It consists of a sequence of commands that specify how a computer should transform input data into output decisions. In organizational contexts, algorithms have become central to coordinating operations, structuring workflows, and enabling autonomous decision-making in increasingly complex environments. Because algorithms are often framed as objective and mathematically grounded, users tend to trust their outputs and adhere to them (Lee et al. 2015).

Algorithmic management refers to a control system in which computer-readable data and software algorithms automate managerial decisions concerning work tasks and workers. Through the delegation of evaluative and supervisory functions to self-learning systems, algorithmic management reduces the need for human oversight in regulating work processes (Duggan et al. 2020; Meijerink and Bondarouk 2023). Conceptually, algorithmic management operates through three structural elements: (1) machine-readable inputs such as behavioral, performance, or transactional data; (2) automated computational processes; and (3) algorithmic outputs in the form of recommendations, rankings, classifications, or decisions.

This structural understanding highlights how algorithmic systems mediate managerial action, serving not only as productivity-enhancing tools but as organizational actors that shape work coordination, evaluation, and control mechanisms.

Table 1 Comparison of Traditional HRM and Algorithmic HRM Processes.

HRM Function	Traditional HRM Process	Algorithmic HRM Process
Workforce Planning	Human managers analyze historical data and rely on intuition for forecasting.	Algorithms predict workforce needs using sales trends, turnover data, and performance analytics.
Recruitment & Selection	Manual resume screening, interviews based on subjective judgment.	AI-based screening, personality/mood analysis, automated interview scheduling and candidate matching.
Training & Development	Uniform training programs designed by HR departments.	Personalized learning paths, algorithmic prediction of skill gaps, adaptive content delivery.
Performance Management	Annual reviews with subjective manager evaluations.	Continuous performance tracking using behavioral data and predictive performance analytics.
Compensation & Rewards	Benchmarking and manual job evaluation.	Real-time salary calibration, performance-linked automated compensation recommendations.
Employee Relations	Surveys and manager intuition to gauge employee sentiment.	Sentiment analysis, turnover prediction, AI-driven promotion or retention interventions.

Types of Algorithms in HRM. Algorithmic HRM systems rely on multiple categories of computational logics, each affecting how HR decisions are generated and justified. Integrating these algorithmic types into a unified framework clarifies the technological foundations of algorithmic HRM.

(1) Deterministic algorithms
Deterministic algorithms follow predefined rules and always produce the same output for a given input (Menard et al. 2023). In HRM, they support routine tasks such as rule-based screening, compliance verification, and workflow routing. Their transparency and predictability make them suitable for administrative and repetitive processes.

(2) Probabilistic algorithms
Probabilistic algorithms generate outputs based on statistical inference, probability distributions, or machine-learning predictions. These systems are fundamental to predictive hiring models, attrition forecasting, and performance risk scoring, where outcomes depend on patterns in large datasets (Fu 2022).

(3) Descriptive analytics
Descriptive analytics summarize existing HR data, identifying patterns in recruitment outcomes, employee performance, turnover behavior, or engagement trends (Cavanagh et al. 2023). While descriptive models do not prescribe actions, they provide diagnostic insights that shape managerial understanding and agenda-setting.

(4) Predictive analytics
Predictive algorithms estimate likely future outcomes such as candidate success probability, turnover likelihood, or performance trajectories using statistical and machine-learning techniques (Alqahtani et al. 2024). These models allow HR departments to anticipate workforce needs, manage risks proactively, and allocate resources strategically.

(5) Normative analytics
Normative algorithms provide prescriptive recommendations, such as optimal staffing decisions, individualized training pathways, or automated compensation suggestions (Rodgers et al. 2023). Representing the highest level of algorithmic influence, normative analytics raise heightened concerns regarding transparency, fairness, and worker autonomy.

These five types illustrate the range of algorithmic logics embedded in HRM systems and establish a coherent technological basis for understanding their organizational implications.

From Traditional HRM to Algorithmic HRM: A Process-Based Comparison. As algorithmic HRM becomes increasingly embedded in organizational systems, it is essential to understand how it reconfigures traditional HR practices. Algorithmic systems not only enhance process efficiency but also fundamentally

reshape the logic, inputs, and outcomes of human resource management. Table 1 below compares traditional HR processes with algorithmic HRM processes across six core HR functions, illustrating how algorithmic systems introduce automation, personalization, and predictive capabilities.

While Table 1 contrasts traditional HRM and algorithmic HRM processes for analytical clarity, this comparison should not be interpreted as a rigid dichotomy. In practice, algorithmic HRM rarely replaces human decision-making entirely. Instead, it reconfigures the distribution of roles between human actors and algorithmic systems. Human judgment, contextual interpretation, and ethical oversight remain integral, particularly in complex or high-stakes HR decisions. Algorithmic HRM thus represents a hybrid governance arrangement in which automation and human discretion coexist, interact, and mutually shape HR practices.

Theoretical Foundations. The growing literature on algorithmic HRM draws on several theoretical streams, yet prior studies often reference these perspectives implicitly without integrating them into a coherent foundation. To situate algorithmic HRM within debates on technology, fairness, and managerial control, this study draws upon three complementary theoretical lenses.

First, the Technology Acceptance Model (TAM) provides a framework for understanding how employees and HR practitioners evaluate and adopt algorithmic systems. Perceived usefulness and perceived ease of use influence individual acceptance of algorithmic tools, shaping whether these systems become integrated into daily HR practices. TAM helps explain variations in employee receptiveness toward automated decision systems and highlights the psychological mechanisms behind technology acceptance in the workplace.

Second, Equity Theory offers insight into how employees interpret the fairness and justice of algorithmic decisions. As algorithmic systems influence hiring, performance evaluation, and promotion, perceptions of distributive, procedural, and interactional justice become central. Equity Theory helps explain why algorithmic opacity, biased data inputs, and perceived discrimination can erode trust and legitimacy, shaping employee reactions to algorithmic governance.

Third, Labor Process Theory provides a sociological foundation for understanding how algorithmic HRM restructures managerial control. Algorithms reconfigure the labor process by redistributing monitoring, evaluation, and supervisory functions from managers to computational systems. This perspective illuminates how algorithmic management reshapes autonomy, resistance, and power relations, thereby linking algorithmic HRM to broader transformations in organizational governance.

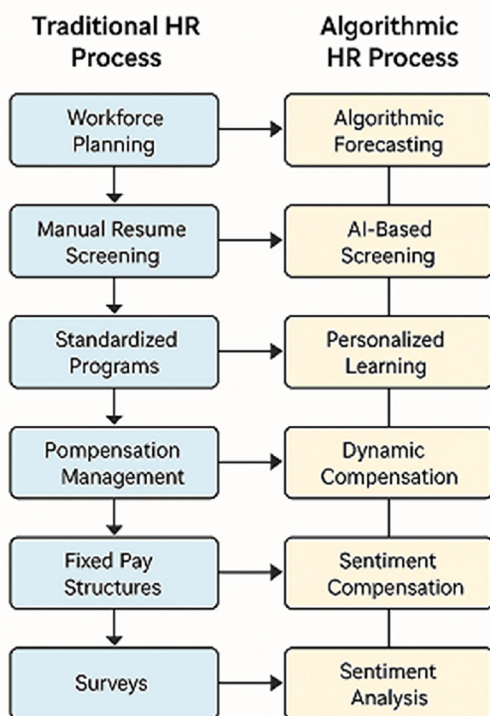


Fig. 1 From Traditional Practices to Intelligent Workflows.

Algorithmic HRM process

The output of algorithm-based strategic HRM is mainly relevant for decision-making in the following sub-areas: HR planning, staffing, training, appraisal, compensation, and employee relations (Leicht-Deobald et al. 2022; Strohmeier 2020). To illustrate the structural transformation brought by algorithmic systems across key HRM domains, the following Fig. 1 compares traditional HR practices with their algorithm-driven counterparts.

The three algorithms relevant to HRM practice include descriptive algorithms that reveal current data patterns and, secondly, predictive algorithms that provide scores representing the probability of outcomes to be used for forecasting purposes. Advanced regression approaches, machine learning algorithms, and methods for data mining could be used (Meijerink and Bondarouk 2023). Finally, the normative algorithms extend the prediction algorithm by using simulation and scenario-based methods to suggest what should be accomplished for possible scenarios (Leicht-Deobald et al. 2022). It has two purposes: (1) decision-making support, where the algorithm proposes scenarios to be followed, and the human managers make the ultimate decision, or (2) decision automation. This study analyzes three algorithms for HRM applications (see Table 2).

Table 2 explains three different algorithms related to HRM. Descriptive algorithms, for example, hiring and selecting (such as filtering CVs), evaluating performance (such as creating multi-source performance ratings), or coaching (e.g., assessing the coaching effectiveness). Predictive algorithms, such as in the areas of hiring and selecting (e.g., forecasting the potential value of jobseekers), human resource planning (e.g., forecasting workforce needs), or performance assessment (e.g., forecasting future employee performance), assist decision-makers. Even if these projections are not the best, they may help HR select an appropriate action course. Normative algorithms, such as software algorithms, filter out candidates based on their CVs, and human managers then decide which candidates to invite for interviews.

Planning. Strategic human resources plans are the starting point for human resources management. Algorithmic technology is vastly superior to humans regarding relevant data analysis, processing, and forecasting. Taking advantage of algorithms in HRM planning will enable organizations to plan their HRM more scientifically. Organizations use algorithmic technology as a decision support system for more comprehensive strategic planning. HRM planning includes taking stock of talent inventory, forecasting workforce needs, gathering talent information, and staff tasking.

(1). Gathering talent information

The data mining method combines information about the world's talent with existing inside and outside talent data. By aggregating this information, a workforce profile is constructed. It allows us to understand the reasonableness of the current HR situation, such as the workforce composition, competitiveness, and skills profile, and to predict, assess, and modify the firm's management in the future. The automated decision-support system's statistics and revision features ultimately provide the organization with the various information it needs.

(2). Talent inventory

Algorithmic technology can update information on talent pool management in real-time, facilitating the organization's talent planning decisions and providing a corresponding talent supply and demand balance solution. The technology can regularly and periodically track the performance of various types of talent, helping to replenish the organization's talent ladder, reasonably evaluate employees' capabilities, and adapt to the needs of corporate talent development.

(3). Forecasting workforce demand

Algorithm technology can predict workforce needs based on sales performance, staff turnover, and future employee performance. Determining future labour demands and making wise recruiting decisions aids in HR planning (Karatop et al. 2015).

(4). Tasking of personnel

Placing the right people in the correct positions is the main challenge in human resources planning. However, algorithmic technology makes this task much easier in the organization. Depending on the employees' documented abilities and performance, the algorithms recommend task assignments to the managers or implement them automatically.

Recruiting. Algorithmic technology assists in human resource planning by identifying future staff demands and making wise hiring decisions (Karatop et al. 2015). Recruitment generally involves screening, scheduling interviews, and an assessment process. The details are as follows.

(1). Screening

Automated content processing software reduces the expense and work involved in CV screening. This technology screens candidates for experience and credentials, manages the hiring manager's recruitment workflow, and helps eliminate bias from job postings (Z. S. Chen 2023). Applicants in the current talent pool are screened against pre-defined parameters, and attributes such as age, sex, marital status, past earnings, and other demographic characteristics are identified. Algorithmic systems can also scan recruitment platforms for possible candidates and automate the analysis of resumes in various formats or languages (Garg et al. 2022).

(2). Scheduling interviews

On the recruitment platform, the algorithm matches clients or requesters with workers and recommends promising candidates to the company recruiter (Carey and Smith 2016). If the recruiter's requirements are met, the algorithmic technology can automatically schedule the candidate for an interview.

Table 2 Three algorithms in HRM.			
	Descriptive algorithms	Predictive algorithms	Normative algorithms
HRM Plan	• Employee profile construction • Employee stock	• Predicting market sales, profits • Predicting future workforce needs	• Automating employee updates • Suggest or automate task assignments
Selection	• CV screening • Assessing candidate personality traits	• Predicting the potential and performance of job seekers • Predicting a job seeker's mood • Predicting candidate engagement	• Automatically matching candidate and job requirements • Automatically recommend candidates for interviews
Training	• Search and collate training materials • Available training plans • Evaluating the efficacy of training	• Forecasting the need for up-skilling • Predicting workforce competency gaps	• Automated coaching for underperforming employees
Compensation	• Online job ranking • Salary calibration • Processing compensation data	• Predicting desired compensation levels	• Automated salary measurement • Automating variable pay
Performance	• Analyze employee behaviour and attitudes • Calculating multi-source performance results	• Predicting task outcomes • Predicting future employee performance	• Automating individual goal-setting • Remind the organization to take corrective action • Automatically warn or suspend under-performing employees
Employee Relations	• Sentiment analysis	• Predicting departures	• Providing managers with recommendations for employee promotion

The following will discuss how the various algorithms are applied explicitly to the sub-functions of HRM.

(3). Assessing candidates

A set of algorithmic applicant screening tools can be integrated into assessing a candidate's process. These screening technologies can be used to identify the personality type of candidates by evaluating their abilities online, assessing how they speak and act to forecast how they will do at work, or using text-mining applications that capture employee sentiment. Their accomplishments, credentials, and competencies are measured concerning the job criteria to assess whether their professional skills match the position's requirements (Faliagka et al. 2014). It may analyze employee-related work behaviour data to predict employee engagement (Al Mehrzi and Singh 2016); Text-mine data may be extracted from employee social media profiles to obtain sentiment analysis (Pitt et al. 2018). These algorithmic systems often utilize natural language processing (NLP) to evaluate job candidates or forecast job results based on written content, micro-expressions, or gaming results. Information about the selected candidate's abilities, personality, skills, dedication, and emotions is used to provide the organization with an objective and unbiased recommendation on the candidate's suitability for the vacancy.

Training. Developing a knowledge base, identifying training needs, planning training, and monitoring training results are the four steps of training enabled by algorithmic systems.

- (1). Building the knowledge base
- This refers to organizational knowledge infrastructure rather than training content itself. Algorithms can automatically web search for relevant knowledge and transmit the knowledge and information to the people who can use it most effectively. The organization's employees can access it by creating a knowledge base and putting it online. Knowledge discovery and data/text mining techniques may be used to inductively identify knowledge relationships and find knowledge trends to develop innovative, organized knowledge systems (Maity 2019). Other methods of knowledge discovery, knowledgeable agents, can be used to aid edge search and retrieval methods for knowledge management systems.
- (2). Mining training needs
- Algorithmic systems can analyze the electronic CV of each current employee, which can help track skill shortages or develop

appropriate training programs. Association rule mining helps automate the analysis of employees' training needs and recommends appropriate courses for them (Chen 2022). These systems might help employees identify their training requirements, create suitable training programs, and finish the required courses if they lack particular abilities. Even underperforming workers can be assisted in determining their training requirements and completing the required courses.

These programs enable individuals to handle their future jobs (Budhwar et al. 2022). Survival analysis methods developed using algorithmic techniques can help organizations predict career levels at different intervals in an employee's career (Garg et al. 2022). Thanks to these projections, employees can benefit from career counselling and proper training throughout their employment with the company.

(3). Training delivery

The course would be finished automatically for staff after they provide their educational goals, profiles, and training points (Bussler and Davis 2002). The daily learning progress of each learner could be tracked using a visual scanning system during training. The typical level of student attendance will then be reliably calculated, and instructional events at various stimulus levels can be retrieved using data analysis.

(4). Training tracking and outcome evaluation

Algorithmic technology can assist in various training facets, including producing high-quality learning summaries and statistical learning analytics. During the training process, Chabot is deployed as a personal career coach for employees, assisting in automatically recording training data, performance evaluation of each participant, and providing helpful remarks and feedback. Organizations may rapidly comprehend the effects of the training by analyzing visual data demonstrating the scope and effectiveness of employee learning (Jia et al. 2018). By assessing training effectiveness and making decisions about employee competencies, the right employees with the appropriate competencies can be assigned to the correct positions (Karatop et al. 2015).

Performance management. Some organizations are moving towards performance management systems that support algorithmic analysis, as advanced technologies produce desirable

employee output behaviours like reduced bias and workplace conflict (Manoharan et al. 2011). Algorithmic involvement in performance assessment includes goal setting, performance evaluation, performance feedback, and improvement.

(1). Goal setting

An algorithmic tool can set individual performance goals for each employee rather than just establishing the same objectives for the same workforce group. It sets goals based on individual past performance, workability, and organizational strategy. They are dynamically tracked and adjusted during the implementation process. Algorithms forecast employee performance levels, relying on their history information and performance attributes (Huang et al. 2006).

(2). Scientific assessment

New applications of algorithms in performance management include utilizing NLP and text analysis to identify subjectivity in the performance review process. It supports organizations to promote employees based on their actual performance rather than considering only years of service (Sivathanu and Pillai 2018). In performance management, algorithmic tools collect and analyze performance information about an employee's work. Established assessment methods like 360-degree feedback take much effort and cost money, but automating algorithmic performance assessments saves expense. Algorithmic techniques analyze and evaluate the full range of individual performance results, supervisor ratings, colleague ratings, and more. These evaluation techniques are designed and input into decision support systems to calculate employee performance from multiple sources efficiently. As a result, algorithmic tools quickly identify underperforming employees based on their combined performance rather than the bias of their superiors.

(3). Performance improvement

Algorithmic techniques can provide continuous performance feedback. Algorithms classify employees based on their productivity and job happiness (Aktepe and Ersoz 2012). With the support of algorithmic techniques, decision-makers can analyze the achievements of individual indicators, suggest shortcomings of failed indicators, develop helpful new metrics, and suggest ways to make improvements. Thus, individual performance improvement plans may be automatically developed based on each participant's skill gaps. By tracking employees' competencies and using data mining techniques, the organization is alerted to improve further employee coaching. It can also alert supervisors to take corrective action for employees, such as providing training and further qualifications (Manoharan et al. 2011). Even the algorithm can warn or stop an employee from working for poor performance.

Reward. The principles, tactics, grades, and structures of compensation packages are determined, allocated, and modified through the monitoring and controlling process known as compensation management (Henderson 2003). Algorithmic techniques can help promote fairness in pay administration, developing pay plans, and handling transactional pay tasks.

(1). Promoting fairness in pay administration

Organizations can build computational rule models that mimic neural information processing. However, such architectures do not inherently guarantee fairness; fairness depends on the quality of input data, bias-mitigation procedures, and governance choices embedded during model design (Abiodun et al. 2018). It is utilized to create a remuneration evaluation system by designing an autonomous decision system.

(2). Developing a pay and benefits plan

The database's skills gap analysis is helpful when establishing the organization's benefits and salary strategies (Pessach et al.

2020). Managers can use algorithmic tools to collect the most pertinent data regarding the necessary employee benefits and remuneration system and make decisions regarding work appraisal.

(3). Assistance with transactional tasks

Algorithmic systems assist practitioners in effectively managing all HR payrolls and calculating and establishing the specifics of workers' labour salaries (Mehrabad and Brojeny 2007). Algorithms also facilitate promotion, maintenance, and compensation scale upgrade decisions (Escolar-Jimenez et al. 2019).

Employee relationship. Algorithm-supported employee relationship management includes employee sentiment analysis, promotion management, and turnover forecasting.

(1). Emotional analysis

Digitized by sensors and intelligent devices, large amounts of digital data from the workplace can be collected in real time to capture employee behaviour, performance, or emotional states (Meijerink & Bondarouk, 2023). Employee thoughts and attitudes in web-based papers, blogs, and social media platforms can be analyzed using text-mining software (Gegenhuber et al. 2021). Using a combination of categorization algorithms and social network analysis, we can analyze team members' emotional highs and lows, the duration of their emotions, and their interaction patterns during their respective team meetings and generate information about the overall team climate.

(2). Promotion management

Algorithm technology can assist in analyzing the flow of talent within a company. By analyzing the performance of employees, the algorithm makes supporting recommendations for promotions for managers to consider.

(3). Turnover forecasting

Managers use algorithms to understand employees' organizational support and job satisfaction and predict employee stability. Algorithms can forecast whether employees are likely to depart the firm so they can conduct retention decisions or estimate the future performance of job candidates to support managers in making selection decisions (Cheng and Hackett 2021). Many criteria, such as job-related ones like promotions, pay increases, and hours worked, or employee-related ones like age, gender, or a combination of both, are considered by algorithms to forecast employee turnover (Rombaut and Guerri 2018). Because of differences in predictors of employee turnover behaviour due to geographical, cultural, and job differences, international companies should assess the appropriateness of different algorithmic turnover models.

Discussion: Algorithmic HRM strengths, weaknesses, and responses

After analyzing the six application areas of algorithmic HRM, this section discusses the implications of algorithmic HRM for organizations, HRM departments, and employees, including strengths, weaknesses, and responses (refer to Table 3).

Top Management and Algorithmic Governance. Algorithmic HRM has been associated with various organizational-level impacts; however, these effects remain contingent on contextual and governance factors.

Algorithmic black box. Algorithmic HRM systems draw on a wide range of input data including behavioral traces (e.g., keystroke logs, productivity metadata, workflow timestamps), historical HR records, communication patterns, and performance archives to generate predictions. Beyond employee-generated data, top management makes design decisions about which variables to

Table 3 Algorithmic HRM strengths, weaknesses, and responses.

	Strength	Weakness	Responses
Organization	• Improve organizational productivity • Reduce costs • Improve operational efficiency • Improve customer engagement and loyalty	• Algorithm Black Box • High technology development cost • Capability gaps	• Improve transparency and interpretability • Build internal data analysis systems in stages to reduce cost pressures • Develop the human capability to run effective algorithmic HRM systems through training
HRM department	• Automate management HRM practices • Eliminates cognitive bias in HRM activities • Reduces administrative tasks for HRM managers	• Accuracy of algorithmic application • Replaces or reshapes positions • Inadequate skills	• Audit of algorithms • A shift in algorithm management to decision support rather than decision-making tools • Develop intuition and empathy skills not available to algorithmic tools • Acquiring integration skills between algorithms and humans
Workers	• Creating favourable employee outcomes • Improving job performance • Developing personal skills • Performing predictable and routine tasks	• Bias • Privacy • Technology use anxiety • Loss of job autonomy	• Use of transparent white-box algorithms • Open communication of information • Protecting privacy through law • Fostering a culture of innovation and training in the use of technology • “Algorithmic activism” resistance strategies

include, how to weigh different features, and which training datasets and optimization criteria to apply. These upstream choices shape outcomes of algorithmic models, yet they are typically opaque to workers and even to HR managers. As a result, organizational actors may experience algorithmic HRM as a “black box,” not only because the computational logic is non-transparent, but also because the design rationales remain invisible (Burrell 2016; Faraj et al. 2018). Many businesses are uncomfortable with the black-box aspect of complicated algorithms when software algorithms are opaque to workers and unaware of the data used by HRM algorithms to generate selected types of output, especially when the suggestions go against what is generally believed (Leicht-Deobald et al. 2022).

We need to improve transparency and interpretability. Actions that re-humanize management decision methods with algorithms ensure clarity in the structure and function of the algorithm, increase reliability and reduce possible biases. Using interpretable artificial intelligence means that algorithmic models give details and justifications for making their functions obvious or understandable (Arrieta et al. 2020). White-box algorithms, which make the algorithm’s design explicit, can be used to develop explicable HRM algorithms (Burrell 2016).

Operational costs. The three main components of AI are algorithms, computers, and large amounts of data (LiT et al. 2019). Nevertheless, creating good AI tools takes better algorithms, necessitating computational processing power and the emergence of supercomputers. As a result, not all organizations can finance such a high level of technology and the expense of burgeoning complex algorithms. The algorithm explanation claims that because businesses can only keep a certain number of archives, it is challenging to train algorithms to function following organizational needs. Organizations can create a strategic vision for the transition of algorithmic HRM by gradually building internal data analysis systems and enhancing information systems to preserve a competitive advantage. The approach will relieve cost pressures, and a database for integrating AI systems will be available (Jia et al. 2018).

HRM department. Algorithmic management automates certain HRM practices. Its application promises to eliminate the racial, gendered, and sexual orientation cognitive biases that taint

decision-making in all HRM functions, reduce the administrative tasks of HR managers, and automate HR decisions (Pessach et al. 2020). Algorithmic management thus results in relatively low marginal and labour costs.

HRM algorithms are adopted to resolve HR-related issues and achieve desired HRM outcomes, such as recruiting the right people, providing personalized training, and dynamically tracking and meeting talent supply (Cheng and Hackett 2021). Therefore, algorithmic management may be thought of as a valuable instrument for HRM that aids in the selection of relevant solutions. (Kellogg et al. 2020).

However, from the HRM department’s perspective, the application of algorithms also needs to consider the application’s accuracy, the risk of job replacement, and skills training.

Accuracy of algorithms application. In the case of fully automated algorithm management, the potential for significant bias in the algorithm can have serious negative consequences for humans (Sienkiewicz 2021). In an organization where demographics dominate, the demographics of the current workforce may be unduly favoured by algorithms that forecast job performance determined by employees’ historical performance. We need to audit the algorithms. It necessitates the creation of fresh organizational HR roles, such as auditing, monitoring, or specification concerning algorithmic HRM.

Analytical applications originating from descriptive, predictive, and narrative sources are attractive to HR professionals, who may be better prepared to apply more automated judgments without considering contextual data and only using the outcomes of algorithmic computations. There could be a propensity to let algorithms make decisions (Sienkiewicz 2021). Such algorithms should not be utilized alone to influence judgments directly. Thus, it is imperative to switch from the administration of algorithms to decision-support tools rather than decision-making mechanisms.

Replacing job positions. By automating the decision-making mechanism, algorithmic human resource management transfers decision-making authority from humans to machines, limiting human involvement and oversight of the labour process. When algorithms automate HR-related decisions, unemployment or de-skilling can occur. The technological deployment of algorithms resonates with the risk of human replacement (Li et al. 2019).

This paradoxical impact of AI on services is examined in Huang and Rust (2018) work substitution theory, which discusses how AI can reshape services and how AI can replace humans altogether, suggesting the development of specific skills for employees that AI cannot yet master intuition and empathy skills. Furthermore, their theory suggests that the role of algorithmic technology can complement rather than compete with people since it facilitates communication with clients or coworkers (Wirtz et al. 2018). While algorithms can do cognitive and analytical activities, high-complexity interpersonal or emotional tasks must be carried out by people or complemented by people and machines (Wirtz 2020).

Inadequate skills. HR managers are underprepared because most lack training in crucial skills and modern technological tools, potentially affecting employees and organizations (Duggan et al. 2020). The challenge for them is to embed new technological skills in their competency suite. To cope with the enhanced role of algorithmic technologies for better organizational outcomes, humans need to develop new skills in an algorithm-driven organizational environment or expand their existing emotional or interpersonal soft skills (Agrawal et al. 2017; Rampersad 2020). To deal with AI algorithms, HR practitioners must learn new abilities, which will help them use their expertise to accomplish various value-added organizational activities. Skills are acquired through retraining and job redesign (Larivière et al. 2017). Wilson and Daugherty (2018) introduce the term “integration skills,” which defines the skills workers should develop to interact effectively with humans and machines. Scholars advocate that organizations and managers should leverage the whole range of potential algorithmic technologies to support the transformation of their enterprise in the following marketing competition by cooperating effectively to exploit market opportunities. The positive effects of technology can be improved when workers are more able to integrate skills to use AI algorithmic technologies.

Workers. Algorithmic technology-centric human resource management creates favourable employee results, including employee satisfaction, dedication, and involvement, thus improving performance (Aouadni and Rebai 2017). Employees can also use AI algorithms to build competencies and develop expected skills (Castellacci and Viñas-Bardolet 2019). In addition, it can perform predictable routine tasks for humans (Maedche et al. 2019). However, algorithmic HRM applications also pose problems of bias, privacy, technology use anxiety, and job autonomy for employees.

Bias. Using historical data sets to train algorithms for talent management can produce discriminatory results. The inherent characteristics of algorithms may make them biased towards particular groups that are disadvantaged or ethically questionable. When algorithmic HRM reinforce discrimination based on gender, race, or age, the final data reflects existing biases against the selected group (Bryce et al. 2022). Since administrators and algorithm designers can determine what can be done with the algorithm, algorithms can also have certain constructed power relations. The algorithm can incorporate arbitrary rules to legitimate some activities while invalidating others, such as collecting personal private information (Meijerink and Bondarouk 2023).

It advocated adopting a transparent white box HRM algorithm to clarify the algorithm’s structure to employees (Burrell 2016). When employees do not understand how the algorithmic system makes decisions, it can lead to hostile organizational behaviour (Tambe et al. 2019). It reduces the perceived fairness of the process that HR algorithms use for assessment purposes, which

can harm organizational commitment. Therefore, employees should be aware of the information being gathered about them and check to see if they have the chance to verify it. Additionally, staff members must comprehend how these algorithm-based choices impact their actions, sentiments, and behaviours (Connelly et al. 2021). People must engage in honest and transparent interactions to guarantee that algorithmic systems operate swiftly, securely, and dependably (Wilson et al. 2017).

Privacy. Algorithms require the input of data that computers can process. Because of the development of mobile technology, employers may now enter information on employee behaviour, emotional state, and social ties into software algorithms (Angrave et al. 2016; Strohmeier 2020). Hacking or unintentional disclosure of this information can compromise people’s privacy. People need laws to protect their privacy from being exploited. For instance, the EU approved the General Data Protection Regulation, which gives individuals increased control over the information and restricts the free flow of the information that corporations gather or handle (Tankard 2016). Data collection related to particular people is protected. Moreover, regulations also apply to algorithms on human resources, requiring employers to reveal to employees the essence of any decisions made by the algorithm, guarantee the right of individuals to contest decisions made only based on the algorithm, and possess the necessary knowledge to tackle the challenges presented by algorithmic decision-making.

Skills gaps. Applying algorithmic technologies to HR involves proliferating technology and integrating the business with strategy, organizational behaviour, corporate culture, and management processes. There is a need to ensure that workers have a sufficient degree of particular competencies before introducing them to teams of human and algorithmic technologies. Therefore, the progressive design and implementation of algorithmic technologies into HR practices and organizational support mechanisms are essential. It can be foreseen that conducting training in algorithmic HRM applications may lead to an attachment relationship between human workers and algorithmic technology. This attachment backbone further enhances the confidence of human workers (Arslan et al. 2022).

Technology application anxiety. Research has found that technological innovations like algorithm management increase employees’ fear of their tasks and jobs (Brougham and Haar 2020). Disappointment, trepidation, and fear are a few of the feelings that represent this anxiousness. In modern firms, human resource departments face great trouble due to this anxiety. Scholars have argued that appropriate and beneficial training is essential to reduce employees’ negative emotions about adapting to new technologies (Bititci et al. 2016). Therefore, creating a new culture and providing technology training can help to overcome technological barriers in the workplace.

Job autonomy. Prior studies also show that algorithms can undertake preparatory or lower-level analytical tasks such as screening, pattern detection, or preliminary scoring, allowing employees to focus on more strategic or conceptual aspects of their jobs. At the same time, the expansion of algorithmic management into scheduling, task allocation, and performance monitoring can gradually narrow employees’ discretion and reshape how autonomy is experienced at work. By automating work processes and embedding evaluation criteria in opaque computational rules, algorithmic management may decrease the perceived liberty and worth of employees (Duggan et al. 2020; Leicht-Deobald et al. 2022), create information asymmetries between workers and organizations, and limit employees’

decision scope (Rosenblat and Stark 2016). Companies may therefore use algorithmic systems as a novel strategy to exert tighter control over workers (Gandini 2019), guiding, evaluating, and training them through interfaces that provide only limited information or recommendations oriented toward the employer's interests (Veen et al. 2020). The opaque nature of algorithmic inputs and outputs reinforces this mode of control and is visible, for example, in algorithm-based HRM roles such as job selection, task assignment, and real-time performance steering.

Employees may react negatively to management decisions. The so-called algorithmic activism involves new resistance strategies such as non-cooperation and providing misleading data to algorithms (Kellogg et al. 2020; Newlands 2021). Non-cooperation leads to neglect of algorithm direction and recommendations. Employees offering fuzzy data to the HRM algorithm to counteract algorithmic control could alter the algorithm's parameters' importance and weight.

Ethical Dilemmas in Algorithmic HRM. Beyond operational risks, algorithmic HRM raises distinct ethical dilemmas that emerge from the governance of data-driven decision systems. These dilemmas can be grouped into four categories, each reflecting a different mechanism through which algorithmic systems reshape the employee relationship.

(1) Data Ethics: Privacy and opaque data practices

Algorithmic HRM depends on continuous data extraction from behavioral, biometric, and digital traces. The core tension arises between efficiency-driven data collection and employees' rights to privacy and informed consent (Capasso et al. 2024). Employers and system designers bear responsibility for implementing transparency protocols, data-minimization standards, and privacy-by-design architectures that comply with regulations such as GDPR and PIPL.

(2) Fairness Ethics: Bias and discrimination

Algorithms may encode or amplify discrimination embedded in historical datasets (Z. Chen 2023). The ethical issue stems from the gap between the perceived neutrality of algorithms and the value-laden nature of training data. HR practitioners and data scientists must monitor disparate impacts through bias audits and fairness-aware machine-learning techniques.

(3) Power Ethics: Delegation and control

The delegation of evaluative tasks to algorithms redistributes power away from employees and line managers (Cristofaro and Bañón-Gomis 2025). The dilemma concerns the erosion of autonomy and the strengthening of managerial control through computational supervision. Governance pathways include human-algorithm co-governance models that clarify decision rights and prevent over-automation.

(4) Responsibility Ethics: Accountability for algorithmic errors

When algorithmic decisions produce harmful outcomes, responsibility becomes diffuse among developers, HR professionals, and platform providers. To address this accountability gap, organizations should document algorithmic decision pathways, define liability rules, and ensure that employees can request explanations or contest decisions (Bartsch et al. 2025).

Limitations and Future Research

This study is conceptual in nature and therefore subject to several limitations that should be acknowledged. First, the analysis does not draw on empirical data to test the causal relationships or relative impact weights of different algorithmic HRM mechanisms. While the framework identifies key domains in which algorithms reshape HR practices, the strength and direction of these effects remain theoretically inferred rather than empirically validated. Future research should employ quantitative surveys,

experiments, or longitudinal designs to examine how specific algorithmic applications influence employee perceptions of fairness, autonomy, and organizational outcomes across contexts.

Second, the applicability of algorithmic HRM is shaped by institutional and regulatory environments, which this study can only address at a general level. Jurisdictions differ markedly in their governance of algorithmic decision-making. For instance, the European Union's General Data Protection Regulation (GDPR) emphasizes algorithmic transparency, data minimization, and the right to contest automated decisions, whereas China's Personal Information Protection Law (PIPL) places stronger emphasis on consent mechanisms, state oversight, and restrictions on data use and transfer. These regulatory differences suggest that algorithmic HRM systems may vary in feasibility, acceptance, and compliance risk across regions. Comparative, cross-national studies are therefore needed to explore how HR algorithms are adapted to distinct legal and institutional settings.

Building on these limitations, several avenues for future research emerge that directly relate to the different HRM activities examined in this study.

First, in HR planning, future studies could investigate how algorithmic forecasting tools interact with strategic workforce planning under different institutional constraints, and how organizations balance human judgment with predictive models when making long-term talent decisions.

Second, in recruitment and selection, researchers may explore how algorithmic screening, interview analytics, and competency-based ranking reshape fairness perceptions, diversity outcomes, and decision accountability. Comparative designs could examine how different algorithmic logics (deterministic versus probabilistic and descriptive versus predictive) produce different candidate evaluation patterns.

Third, in training and development, an important direction concerns how algorithmic learning systems construct individualized training pathways. Future research can examine the socio-material practices of how HR professionals curate training data, how employees appropriate algorithmic recommendations, and how knowledge infrastructures influence the formation of digital skills.

Fourth, in performance management, scholars may study how algorithmic scoring, behavioral monitoring, and real-time feedback influence employee autonomy, task motivation, and the dynamics of algorithmic surveillance. Experimental and longitudinal work could assess how algorithm-based evaluations differ from human evaluations in terms of accuracy, perceived justice, and behavioral consequences.

Fifth, in reward and compensation, future studies could analyze the implications of algorithmic pay-setting, incentive allocation, and performance-linked reward optimization. In particular, researchers may explore whether algorithmic compensation models reinforce or reduce systemic inequalities, and how transparency mechanisms affect employee acceptance.

Across these domains, future research should also draw on the broader literature on algorithmic governance, power asymmetries, and human-algorithm interaction to build integrative theories that capture how algorithms reshape managerial rationality in HRM. Multi-method designs combining ethnography, experiments, and computational audits would further deepen understanding of algorithmic HRM mechanisms.

In addition to these domain-specific themes, future research should also adopt diverse methodological approaches. Quantitative surveys and experimental designs could help identify causal mechanisms underlying employee reactions to algorithmic decision-making. Comparative case studies across organizations and industries may illuminate the contextual contingencies that shape algorithmic legitimacy. Mixed-method designs, combining

digital trace data with interviews or ethnographic observations, would enrich our understanding of how algorithmic systems interact with different employment arrangements.

Practical Implications

The findings of this study offer several practical implications for organizations seeking to implement algorithmic HRM responsibly. First, organizations should align the adoption of algorithmic HRM with their broader strategic objectives rather than treating algorithmic tools as isolated technological solutions. When organizational goals emphasize sustainability, fairness, and long-term capability development, algorithmic engagement can support more consistent and transparent HR practices, provided that human oversight is maintained.

Second, the design and implementation of algorithmic HRM systems should be approached as an iterative and participatory process. Organizations are encouraged to explore the links between strategic priorities, HR objectives, and algorithmic applications through pilot projects, algorithmic modelling, and mixed-method evaluations. Qualitative assessments involving managers and employees can help identify unintended consequences and refine system design before large-scale deployment.

Finally, organizations should consider the organizational readiness required for algorithmic HRM. This includes investing in data governance capabilities, clarifying accountability for algorithmic decisions, and developing employee competencies for working alongside algorithmic systems. When algorithmic HRM initiatives are introduced transparently and in the interests of multiple stakeholders, organizations are better positioned to scale these systems in ways that enhance efficiency while preserving trust, fairness, and human agency.

Conclusion

The study concludes that emerging rapid technological developments within the backdrop of the fourth industrial revolution provide an intelligent digital background for HRM practices. HRM responsibilities, such as HR planning, recruitment, training, compensation, performance management, and employee relations, can be influenced by algorithmic management. Algorithmic HRM is considered to increase HR-related decision-making quality and automate HRM activities through software based on digital data. Beyond these practical implications, the analysis also shows that the three theoretical perspectives mobilized in this study (the Technology Acceptance Model, Equity Theory, and Labor Process Theory) offer complementary lenses for understanding algorithmic HRM. TAM explains how employees and HR practitioners evaluate and adopt algorithmic systems, Equity Theory clarifies how fairness perceptions and justice concerns shape the legitimacy of algorithmic decisions, and Labour Process Theory highlights how algorithmic management reconfigures control, autonomy, and power relations in the workplace. Taken together, these perspectives suggest that algorithmic HRM should be understood not only as a technological innovation but as a mode of algorithmic governance that reshapes managerial rationality and the employment relationship. We therefore need to look at algorithmic HRM across different disciplinary boundaries. These dual effects underline the need for context-aware implementation and robust governance. The study recommends that organizations take countermeasures to address the weaknesses of algorithmic HRM in the application of the system.

Data availability

No datasets were generated or analyzed during this study. The manuscript is purely conceptual, and therefore no data are associated with the work.

Received: 29 October 2025; Accepted: 7 March 2026;

Published online: 16 March 2026

References

- Abiodun OI, Jantan A, Omolara AE, Dada KV, Mohamed NA, Arshad H (2018) State-of-the-art in artificial neural network applications: A survey. *Heliyon* 4(11):e00938
- Abraham M, Niessen C, Schnabel C, Lorek K, Grimm V, Möslin K, Wrede M (2019) Electronic monitoring at work: The role of attitudes, functions, and perceived control for the acceptance of tracking technologies. *Hum Resour Manag J* 29(4):657–675
- Agrawal A, Gans J, Goldfarb A (2017) What to expect from artificial intelligence. MIT Sloan Management Review, Cambridge, MA, USA
- Aktepe A, Ersoz S (2012) A Quantitative Performance Evaluation Model Based on a Job Satisfaction-Performance Matrix and Application in a Manufacturing Company. *Int J Industr Eng* 19(6):264–277
- Al Mehrzi N, Singh SK (2016) Competing through employee engagement: a proposed framework. *Int J Product Perform Manag* 65(6):831–843
- Alqahtani H, Almagrabi H, Alharbi A (2024) Employee attrition prediction using machine learning models: A review paper *Int J Art Intell Appl* 15(2):23–49
- Angrave D, Charwood A, Kirkpatrick I, Lawrence M, Stuart M (2016) HR and analytics: Why HR is set to fail the big data challenge. *Hum Resour Manag J* 26(1):1–11
- Aouadni I, Rebai A (2017) Decision support system based on genetic algorithm and multi-criteria satisfaction analysis (MUSA) method for measuring job satisfaction. *Ann Oper Res* 256:3–20
- Arrieta AB, Diaz-Rodríguez N, Del Ser J, Benetot A, Tabik S, Barbado A, García S, Gil-López S, Molina D, Benjamins R (2020) Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf fusion* 58:82–115
- Arslan A, Cooper C, Khan Z, Gölgeci I, Ali I (2022) Artificial intelligence and human workers interaction at team level: a conceptual assessment of the challenges and potential HRM strategies. *Int J Manpow* 43(1):75–88
- Bartsch SC, Nguyen LH, Schmidt J-H, Du G, Adam M, Benlian A, Sunyaev A (2025) The Present and Future of Accountability for AI Systems: A Bibliometric Analysis. *Inform Syst Front* 27:1–22
- Bititci U, Cocca P, Ates A (2016) Impact of visual performance management systems on the performance management practices of organisations. *Int J Prod Res* 54(6):1571–1593
- Bondarouk T, Parry E, Furmueller E (2017) Electronic HRM: four decades of research on adoption and consequences. *Int J Hum Resour Manag* 28(1):98–131
- Brougham D, Haar J (2020) Technological disruption and employment: The influence on job insecurity and turnover intentions: A multi-country study. *Technol Forecast Soc Change* 161:120276
- Bryce V, Brooks L, Stahl B (2022) We need to talk about digital HR ethics! A review of the academic literature on ethical aspects of algorithmic Human Resource Management (HRM) technologies. Available at: <https://www.researchgate.net/profile/Vincent-Bryce/publication/357900086>
- Budhwar P, Malik A, De Silva MTT, Thevisuthan P (2022) Artificial intelligence—challenges and opportunities for international HRM: a review and research agenda. *Int J Hum Resour Manag* 33(6):1065–1097
- Burrell J (2016) How the machine ‘thinks’: Understanding opacity in machine learning algorithms. *Big data Soc* 3(1):2053951715622512
- Bussler L, Davis E (2002) Information systems: The quiet revolution in human resource management. *J computer Inf Syst* 42(2):17–20
- Capasso M, Arora P, Sharma D, Tacconi C (2024) On the right to work in the age of artificial intelligence: Ethical safeguards in algorithmic human resource management. *Bus Hum Rights J* 9(3):346–360
- Carey D, Smith M (2016) How companies are using simulations, competitions, and analytics to hire. *Harv Bus Rev* 22(2):32–56
- Castellacci F, Viñas-Bardolet C (2019) Internet use and job satisfaction. *Computers Hum Behav* 90:141–152
- Cavanagh J, Pariona-Cabrera P, Halvorsen B (2023) In what ways are HR analytics and artificial intelligence transforming the healthcare sector? *Asia Pac J Hum Resour* 61(4):785–793
- Chen Z (2022) Artificial intelligence-virtual trainer: Innovative didactics aimed at personalized training needs. *J Knowledge Econ* 14(2):1–19
- Chen Z (2023) Ethics and discrimination in artificial intelligence-enabled recruitment practices. *Humanities Soc Sci Commun* 10(1):1–12
- Chen ZS (2023) Collaboration among recruiters and artificial intelligence: removing human prejudices in employment. *Cognition, Technol Work* 25(1):135–149
- Cheng MM, Hackett RD (2021) A critical review of algorithms in HRM: Definition, theory, and practice. *Hum Resour Manag Rev* 31(1):100698

- Connelly CE, Fieseler C, Černe M, Giessner SR, Wong SI (2021) Working in the digitized economy: HRM theory & practice. *Hum Resour Manag Rev* 31(1):100762
- Cristofaro M, Bañón-Gomis AJ (2025) Dancing with the algorithm: a framework to navigate knowledge and autonomy in AI-assisted managerial decisions. *J Knowledge Manag* 30:1–31
- Duggan J, Sherman U, Carbery R, McDonnell A (2020) Algorithmic management and app-work in the gig economy: A research agenda for employment relations and HRM. *Hum Resour Manag J* 30(1):114–132
- Escolar-Jimenez CC, Matsuzaki K, Okada K, Gustilo RC (2019) Data-driven decisions in employee compensation utilizing a neuro-fuzzy inference system. *Int J Emerg Trends Eng Res* 7(8):163
- Faliagka E, Iliadis L, Karydis I, Rigou M, Sioutas S, Tsakalidis A, Tzimas G (2014) On-line consistent ranking on e-recruitment: seeking the truth behind a well-formed CV. *Artif Intell Rev* 42:515–528
- Faraj S, Pachidi S, Sayegh K (2018) Working and organizing in the age of the learning algorithm. *Inf Organ* 28(1):62–70
- Fu X (2022) Statistical machine learning model for capacitor planning considering uncertainties in photovoltaic power. *Prot Control Mod Power Syst* 7(1):1–13
- Gandini A (2019) Labour process theory and the gig economy. *Hum Relat* 72(6):1039–1056
- Garcia-Arroyo J, Osca A (2021) Big data contributions to human resource management: a systematic review. *Int J Hum Resour Manag* 32(20):4337–4362
- Garg S, Sinha S, Kar AK, Mani M (2022) A review of machine learning applications in human resource management. *Int J Product Perform Manag* 71(5):1590–1610
- Gegenhuber T, Ellmer M, Schüßler E (2021) Microphones, not megaphones: Functional crowdworker voice regimes on digital work platforms. *Hum Relat* 74(9):1473–1503
- Henderson RI (2003) Compensation management in a knowledge-based world. Prentice Hall
- Huang M-H, Rust RT (2018) Artificial intelligence in service. *J Serv Res* 21(2):155–172
- Huang M-J, Tsou Y-L, Lee S-C (2006) Integrating fuzzy data mining and fuzzy artificial neural networks for discovering implicit knowledge. *Knowl-Based Syst* 19(6):396–403
- Jia Q, Guo Y, Li R, Li Y, Chen Y (2018) A conceptual artificial intelligence application framework in human resource management. *Proceedings of the International Conference on Electronic Business*
- Karatop B, Kubat C, Uygün Ö (2015) Talent management in manufacturing system using fuzzy logic approach. *Computers Ind Eng* 86:127–136
- Kellogg KC, Valentine MA, Christin A (2020) Algorithms at work: The new contested terrain of control. *Acad Manag Ann* 14(1):366–410
- Larivière B, Bowen D, Andreassen TW, Kunz W, Sirianni NJ, Voss C, Wunderlich NV, De Keyser A (2017) “Service Encounter 2.0”: An investigation into the roles of technology, employees and customers. *J Bus Res* 79:238–246
- Lee MK, Kusbit D, Metsky E, Dabbish L (2015) Working with machines: The impact of algorithmic and data-driven management on human workers
- Leicht-Deobald U, Busch T, Schank C, Weibel A, Schafheitle S, Wildhaber I, Kasper G (2022) The challenges of algorithm-based HR decision-making for personal integrity. In *Business and the Ethical Implications of Technology* (pp. 71–86). Springer
- Li JJ, Bonn MA, Ye BH (2019) Hotel employee’s artificial intelligence and robotics awareness and its impact on turnover intention: The moderating roles of perceived organizational support and competitive psychological climate. *Tour Manag* 73:172–181
- Li T, Wang T, Yuan B (2019) New Module Research of Enterprises’ Training and Development Based on AI Technology. *Mod Shoppingmall* 000(008):75–76
- Maedche A, Legner C, Benlian A, Berger B, Gimpel H, Hess T, Hinz O, Morana S, Sollner M (2019) AI-based digital assistants. *Bus Inf Syst Eng* 61(4):535–544. In
- Maity S (2019) Identifying opportunities for artificial intelligence in the evolution of training and development practices. *J Manag Dev* 38(8):12–32
- Manoharan TR, Muralidharan C, Deshmukh SG (2011) An integrated fuzzy multi-attribute decision-making model for employees’ performance appraisal. *Int J Hum Resour Manag* 22(03):722–745
- Marler JH, Boudreau JW (2017) An evidence-based review of HR Analytics. *Int J Hum Resour Manag* 28(1):3–26
- Mehrabad MS, Brojeny MF (2007) The development of an expert system for effective selection and appointment of the jobs applicants in human resource management. *Computers Ind Eng* 53(2):306–312
- Meijerink J, Bondarouk T (2023) The duality of algorithmic management: Toward a research agenda on HRM algorithms, autonomy and value creation. *Hum Resour Manag Rev* 33(1):100876
- Menard C, Lohstroh M, Bateni S, Chorlian M, Deng A, Donovan P, Fournier C, Lin S, Suchert F, Tanneberger T (2023) High-performance deterministic concurrency using lingua franca. *ACM Trans Architecture Code Optim* 20(4):1–29
- Newlands G (2021) Algorithmic surveillance in the gig economy: The organization of work through Lefebvrian conceived space. *Organ Stud* 42(5):719–737
- Newman DT, Fast NJ, Harmon DJ (2020) When eliminating bias isn’t fair: Algorithmic reductionism and procedural justice in human resource decisions. *Organ Behav Hum Decis Process* 160:149–167
- Pessach D, Singer G, Avrahami D, Ben-Gal HC, Shmueli E, Ben-Gal I (2020) Employees recruitment: A prescriptive analytics approach via machine learning and mathematical programming. *Decis Support Syst* 134:113290
- Pitt CS, Botha E, Ferreira JJ, Kietzmann J (2018) Employee brand engagement on social media: Managing optimism and commonality. *Bus Horiz* 61(4):635–642
- Rampersad G (2020) Robot will take your job: Innovation for an era of artificial intelligence. *J Bus Res* 116:68–74
- Rodgers W, Murray JM, Stefanidis A, Degbey WY, Tarba SY (2023) An artificial intelligence algorithmic approach to ethical decision-making in human resource management processes. *Hum Resour Manag Rev* 33(1):100925
- Rombaut E, Guerry M-A (2018) Predicting voluntary turnover through human resources database analysis. *Manag Res Rev* 41(1):96–112
- Rosenblat A, Stark L (2016) Algorithmic labor and information asymmetries: A case study of Uber’s drivers. *Int J Commun* 10:27
- Shank DB, Graves C, Gott A, Gamez P, Rodriguez S (2019) Feeling our way to machine minds: People’s emotions when perceiving mind in artificial intelligence. *Computers Hum Behav* 98:256–266
- Sienkiewicz Ł (2021) Algorithmic Human Resources Management—Perspectives and Challenges. *Annales Universitatis Mariae Curie-Skłodowska, Sect H Oeconomia* 55(2):95–105
- Sivathanu B, Pillai R (2018) Smart HR 4.0—how industry 4.0 is disrupting HR. *Hum Resour Manag Int Dig* 26(4):7–11
- Strohmeier S (2020) Smart HRM—a Delphi study on the application and consequences of the Internet of Things in Human Resource Management. *Int J Hum Resour Manag* 31(18):2289–2318
- Tambe P, Cappelli P, Yakubovich V (2019) Artificial intelligence in human resources management: Challenges and a path forward. *Calif Manag Rev* 61(4):15–42
- Tankard C (2016) What the GDPR means for businesses. *Netw Security* 2016(6):5–8
- Torres EN, Mejia C (2017) Asynchronous video interviews in the hospitality industry: Considerations for virtual employee selection. *Int J Hospitality Manag* 61:4–13
- Vassilopoulou J, Kyriakidou O, Özbilgin MF, Groutsis D (2022) Scientism as illusio in HR algorithms: Towards a framework for algorithmic hygiene for bias proofing. *Hum Res Manag J* 34(2):311–325
- Veen A, Barratt T, Goods C (2020) Platform-capital’s ‘app-etite’ for control: A labour process analysis of food-delivery work in Australia. *Work, Employ Soc* 34(3):388–406
- Wilson HJ, Daugherty P, Bianzino N (2017) The jobs that artificial intelligence will create. *MIT Sloan Manag Rev* 58(4):14
- Wilson HJ, Daugherty PR (2018) Collaborative intelligence: Humans and AI are joining forces. *Harv Bus Rev* 96(4):114–123
- Wirtz J (2020) Organizational ambidexterity: cost-effective service excellence, service robots, and artificial intelligence. *Organ Dyn* 49(3):1–9
- Wirtz J, Patterson PG, Kunz WH, Gruber T, Lu VN, Paluch S, Aarts G, Martins A (2018) Brave new world: service robots in the frontline. *J Serv Manag* 29(5):907–931
- Yaniv E (2011) Construct clarity in theories of management and organization. *Acad Manag Rev* 36(3):590–592
- Zhang Y, Xu S, Zhang L, Yang M (2021) Big data and human resource management research: An integrative review and new directions for future research. *J Bus Res* 133:34–50

Acknowledgements

Portions of the language polishing during the revision process were assisted by ChatGPT. All conceptual development, arguments, interpretations, and final revisions were solely performed by the author. This research received no specific grant from any funding agency, commercial, or not-for-profit sectors.

Author contributions

Zhisheng Chen solely conceived the study, developed the conceptual framework, conducted the literature analysis, drafted all sections of the manuscript, and completed all revisions in response to reviewer comments. Language polishing during revision was partially supported by ChatGPT, but all intellectual contributions, arguments, and interpretations were developed by the author.

Competing interests

The author declares no competing interests.

Ethical approval

Ethical approval was not required for this study because it is a conceptual analysis that does not involve human participants, personal data, or any experimental intervention.

Informed consent

Informed consent was not required, as the study does not collect or analyze data from human participants.

Additional information

Correspondence and requests for materials should be addressed to Zhisheng Chen.

Reprints and permission information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026